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SECTION :- 4A

COURSE :- Probability & Random Processes

Problem # 1

Let W be the random variable giving the number of head minus the number of tails in 3 tosses of a coin. List elements of sample space S for 3 tosses assign a value of W

Solution :-

We will denote the elements of sample space S with a sequence of 3 letters, each of which will be either H or T

In the k -th toss, if the coin lands on the head, there will be letter H on the k -th position in the letter sequence. If the coin lands on the tails, letter T will occupy the k -th position in the letter sequence, $k \in \{1, 2, 3\}$

Therefore, elements of the sample space & their assigned values of the random variable W are :

$$HHH \rightarrow W = 3$$

$$HHT \rightarrow W = 1$$

$$HTH \rightarrow W = 1$$

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$$THH \rightarrow W = 1$$

$$HTT \rightarrow W = 1$$

$$THT \rightarrow W = -1$$

$$TTH \rightarrow W = -1$$

$$TTT \rightarrow W = -3$$

We obtain the value of W by subtracting the number of T-s from the number of letters H in each element of the sample space S .

Problem #2

A coin is flipped until 3 heads in succession occur. List only those elements of sample space that require 6 or less tosses. Is this a discrete sample space.

Solution:-

$$S = \{ HHH, THHH, TTHHH, TTTHHH, \dots, HTHHH, HTTHHH, HTTTHHH, \dots \}$$

List of sample space that requires 6 or less tosses is

$$L = \{ HHH, THHH, TTHHH, TTTHHH, HTHHH, HTTHHH, HHH, THHH, THTHHH \}$$

The sample space contains an unending sequence with as many elements as there are whole numbers, so the sample space is discrete.

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Problem #3

$$F(t) = \begin{cases} 0 & t < 1 \\ 1/4 & 1 \leq t < 3 \\ 1/2 & 3 \leq t < 5 \\ 3/4 & 5 \leq t < 7 \\ 1 & t \geq 7 \end{cases}$$

(a) $P(T = 5)$

$$P(T = 5) = P(T \leq 5) - P(T < 5)$$

(1) $F(5) - \lim_{t \rightarrow 5^-} P(T \leq t)$

$$= F(5) - \lim_{t \rightarrow 5^-} F(t)$$

(2) $F(5) - \lim_{t \rightarrow 5^-} \frac{1}{2}$

$$= \frac{3}{4} - \frac{1}{2}$$

$$= \frac{1}{4}$$

In (1) we used the definition of the cumulative distribution function of the random variables T :

$$F(t) = P(T \leq t)$$

We have also written $P(T < 5)$ as $\lim_{t \rightarrow 5^-} P(T \leq t)$

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In (2) we used the fact
 $F(t) = \frac{1}{2}$ for every t such that $3 \leq t < 5$

(b) $P(T > 3)$;

$$\begin{aligned} P(T > 3) &= 1 - P(T \leq 3) \\ &= 1 - F(3) \\ &= 1 - \frac{1}{2} \\ &= \frac{1}{2} \end{aligned}$$

In the first equality, we have rewritten the expression in such manner because

$$\begin{aligned} 1 &= P(T \in \mathbb{R}) \\ &= P(T \in (-\infty, 3] \cup (3, \infty)) \\ &= P(T \leq 3) + P(T > 3) \end{aligned}$$

(c) $P(1.4 < T < 6)$;

$$\begin{aligned} P(1.4 < T < 6) &= P(T < 6) - P(T \leq 1.4) \\ &= \lim_{t \rightarrow 6^-} P(T \leq t) - P(T \leq 1.4) \\ &= \lim_{t \rightarrow 6^-} F(t) - F(1.4) \end{aligned}$$

$$(1) \quad \lim_{t \rightarrow 6^-} \frac{3}{4} - \frac{1}{4}$$

$$= \frac{3}{4} - \frac{1}{4}$$

$$= \frac{1}{2}$$

$F(t) = 3/4$ for every t such that $5 \leq t < 7$

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(d) $P(T \leq 5 | T \geq 2);$

$$P(T \leq 5 | T \geq 2) = \frac{P(T \leq 5 \cap T \geq 2)}{P(T \geq 2)} \quad \text{--- ①}$$

$$= \frac{P(2 \leq T \leq 5)}{1 - P(T < 2)}$$

$$= \frac{P(T \leq 5) - P(T < 2)}{1 - P(T < 2)}$$

$$= \frac{F(5) - \lim_{t \rightarrow 2^-} P(T \leq t)}{1 - \lim_{t \rightarrow 2^-} P(T \leq t)}$$

$$= \frac{F(5) - \lim_{t \rightarrow 2^-} F(t)}{1 - \lim_{t \rightarrow 2^-} F(t)}$$

$$= \frac{3/4 - \lim_{t \rightarrow 2^-} 1/4}{1 - \lim_{t \rightarrow 2^-} 1/4} \quad \text{--- ②}$$

$$= \frac{3/4 - 1/4}{1 - 1/4}$$

$$= \frac{1/2}{3/4}$$

$$= 2/3$$

In ① we used the formula for conditional probability

In ② we used the fact

$$F(t) = \frac{1}{4} \text{ for every } t \text{ such that } 1 \leq t < 3$$

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Problem #4

x	0	1	2	3	4
$f(x)$	0.41	0.37	0.16	0.05	0.01

Construct commulative distribution function of X .

Solution:-

$$F(x) = \sum_{t \leq x} f(t)$$

$$F(0) = \sum_{t \leq 0} f(t) = f(0) = 0.41$$

$$F(1) = \sum_{t \leq 1} f(t) = f(0) + f(1) = 0.41 + 0.37 = 0.78$$

$$F(2) = \sum_{t \leq 2} f(t) = f(0) + f(1) + f(2) = 0.41 + 0.37 + 0.16 = 0.94$$

$$F(3) = \sum_{t \leq 3} f(t) = f(0) + f(1) + f(2) + f(3) = 0.41 + 0.37 + 0.16 + 0.05$$

$$F(3) = 0.99$$

$$F(4) = \sum_{t \leq 4} f(t) = f(0) + f(1) + f(2) + f(3) + f(4)$$

$$= 0.41 + 0.37 + 0.16 + 0.05 + 0.01 = 1$$

x	0	1	2	3	4
$F(x)$	0.41	0.78	0.94	0.99	1

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Problem #5

$$f(x) = \begin{cases} k\sqrt{x}, & 0 < x < 1 \\ 0, & \text{elsewhere} \end{cases}$$

Solution:-

(a) Evaluate k

$$\int_{-\infty}^{\infty} f(x) dx = 1$$

$$1 = \int_{-\infty}^{\infty} f(x) dx$$

$$= \int_0^1 k\sqrt{x} dx$$

$$= k \int_0^1 \sqrt{x} dx$$

$$= k \left(\frac{2}{3} \sqrt{x^3} \right) \Big|_0^1$$

$$= \frac{2k}{3} \cdot (1-0)$$

$$= 2k/3$$

$$\boxed{k = 3/2}$$

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(b) Find $F(x)$ and use it to evaluate
 $P(0.3 < X < 0.6)$

$$F(x) = P(X \leq x) = \int_{-\infty}^x f(x) dx \quad \text{--- (1)}$$

The equality in 1 hold because f is 0 for $x \leq 0$ and the integral of the zero-function over a segment is equal to 0

Because of it we also conclude

$$F(x) = 0 \quad \text{for } x \leq 0.$$

Furthermore, for $0 < x < 1$ we have

$$F(x) = \int_0^x f(x) dx$$

$$= \int_0^x \frac{3}{2} \sqrt{x} dx$$

$$= \frac{3}{2} \cdot \left(\frac{2}{3} \sqrt{x^3} \right) \Big|_0^x$$

$$= \sqrt{x^3} - 0$$

$$= \sqrt{x^3}$$

Finally for $x > 1$ we know that

$$F(x) = \int_0^x f(x) dx = \int_0^1 f(x) dx$$

since f is a zero function for $x \geq 1$ and so its integral over the segment $[1, x]$ equals 0.

But from (a) we know that $\int_0^1 f(x) dx = 1$ so obviously

$$F(x) = 1 \quad \text{for } x \geq 1$$

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Now, we can use the cumulative distribution function in order to determine

$$\begin{aligned}P(0.3 < X < 0.6) &= P(X < 0.6) - P(X \leq 0.3) \\&= P(X \leq 0.6) - P(X \leq 0.3) \\&= F(0.6) - F(0.3) \\&= \sqrt{(0.6)^3} - \sqrt{(0.3)^3}\end{aligned}$$

$$P(0.3 < X < 0.6) = 0.3004$$

Problem #6

$$f(x) = \begin{cases} 2(1-x) & 0 < x < 1 \\ 0, & \text{otherwise} \end{cases}$$

(a) Calculate $P(X \leq 1/3)$

$$\begin{aligned}P(X < 1/3) &= \int_{-\infty}^{1/3} f(x) dx \\&= \int_0^{1/3} 2(1-x) dx \\&= 2 \left(x - \frac{x^2}{2} \right) \Big|_0^{1/3} \\&= 2 \left[\frac{1}{3} - \left(\frac{1}{3} \right)^2 \cdot \frac{1}{2} - 0 + \frac{0^2}{2} \right] \\&= 2 \left(\frac{1}{3} - \frac{1}{18} \right)\end{aligned}$$

$$P(X < 1/3) = 5/9$$

(b) What is the probability that X will exceed 0.5?

$$P(X > 0.5) = P(0.5 < X < 1) \quad \text{--- ①}$$

$$= \int_{0.5}^1 f(x) dx$$

$$= \int_{0.5}^1 2(1-x) dx$$

$$= 2 \left(x - \frac{x^2}{2} \right) \Big|_{0.5}^1$$

$$= 2 \left(1 - \frac{1^2}{2} - 0.5 + \frac{0.5^2}{2} \right)$$

$$= (2)(1/8)$$

$$P(X > 0.5) = 1/4$$

The equality in ① holds because X is a random variable which assumes only the values between 0 and 1, which means that $P(X > 1) = 0$

(c) Given that $X \geq 0.5$, what is the probability that X will be less than 0.75?

$$P(X < 0.75 | X \geq 0.5) = \frac{P[(X < 0.75) \cap (X \geq 0.5)]}{P(X \geq 0.5)}$$

$$= \frac{P(0.5 < X < 0.75)}{P(0.5 \leq X \leq 1)}$$

$$= \frac{\int_{0.5}^{0.75} f(x) dx}{\int_{0.5}^1 f(x) dx}$$

$$= \frac{\int_{0.5}^{0.75} 2(1-x) dx}{\int_{0.5}^1 2(1-x) dx}$$

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$$\begin{aligned}
 &= \frac{\int_{0.5}^{0.75} 2(1-x) dx}{\int_{0.5}^{0.75} 2(1-x) dx} \\
 &= \frac{2 \left(x - \frac{x^2}{2} \right) \Big|_{0.5}^{0.75}}{2 \left(x - \frac{x^2}{2} \right) \Big|_{0.5}^{0.75}} \\
 &= \frac{0.75 - \frac{0.75^2}{2} - 0.5 + \frac{0.5^2}{2}}{1 - \frac{1^2}{2} - 0.5 + \frac{0.5^2}{2}} \\
 &= 3/32 / 1/8 \\
 &= 3/4
 \end{aligned}$$

Problem # 7

If the joint probability distribution of X & Y is given by

$$f(x, y) = \frac{x+y}{30}, \text{ for } x=0, 1, 2, 3; y=0, 1, 2;$$

(a) $P(X \leq 2, Y=1)$

$$\begin{aligned}
 P(X \leq 2, Y=1) &= P(X=0, Y=1) + P(X=1, Y=1) + P(X=2, Y=1) \\
 &= f(0, 1) + f(1, 1) + f(2, 1) \\
 &= \frac{0+1}{30} + \frac{1+1}{30} + \frac{2+1}{30} \\
 &= 6/30 \\
 &= 1/5
 \end{aligned}$$

(b) $P(X \leq 2, Y = 1);$

$$\begin{aligned}
 P(X \leq 2, Y = 1) &= P(X \geq 3, Y \leq 1) \\
 &= P(X = 3, Y \leq 1) \\
 &= P(X = 3, Y = 0) + P(X = 3, Y = 1) \\
 &= f(3, 0) + f(3, 1) \\
 &= \frac{3+0}{30} + \frac{3+1}{30} \\
 &= 7/30
 \end{aligned}$$

(c) $P(X > Y);$

$$\begin{aligned}
 P(X > Y) &= \sum_{y=0}^2 \sum_{x=y+1}^3 P(X=x, Y=y) \\
 &= \left(\frac{1+0}{30} + \frac{2+0}{30} + \frac{3+0}{30} \right) + \left(\frac{2+1}{30} + \frac{3+1}{30} \right) + \left(\frac{3+2}{30} \right) \\
 &= \frac{6}{30} + \frac{7}{30} + \frac{5}{30} \\
 &= 3/5
 \end{aligned}$$

(d) $P(X+Y = 4)$

$$\begin{aligned}
 P(X+Y=4) &= \sum_{y=0}^2 P(X=4-y, Y=y) \\
 &= P(X=4, Y=0) + P(X=3, Y=1) + P(X=2, Y=2) \\
 &= 0 + f(3, 1) + f(2, 2) \quad \text{--- ①} \\
 &= \frac{3+1}{30} + \frac{2+2}{30} \\
 &= 4/15
 \end{aligned}$$

Problem #8

$$f(x,y) = \begin{cases} \frac{2}{3}(x+2y) & , 0 \leq x \leq 1, 0 \leq y \leq 1, \\ 0 & , \text{elsewhere} \end{cases}$$

(a) Find the marginal density of X

$$\begin{aligned} g(x) &= \int_{-\infty}^{\infty} f(x,y) dy \\ &= \int_0^1 \frac{2}{3}(x+2y) dy \\ &= \frac{2}{3}(xy + y^2) \Big|_0^1 \\ &= \frac{2}{3}[(x+1^2) - (0 \cdot x + 0^2)] \\ &= \frac{2}{3}(x+1) \end{aligned}$$

(b) Find the marginal density of Y

$$\begin{aligned} h(y) &= \int_{-\infty}^{\infty} f(x,y) dx \\ &= \int_0^1 \frac{2}{3}(x+2y) dx \\ &= \frac{2}{3}\left(\frac{x^2}{2} + 2xy\right) \Big|_0^1 \\ &= \frac{2}{3}\left[\left(\frac{1^2}{2} + 2y\right) - \left(\frac{0^2}{2} + 2y \cdot 0\right)\right] \\ &= \frac{1}{3}(4y+1) \end{aligned}$$

- (c) Find the probability that the drive-through facility is busy less than one-half of the time

$$P(X < 0.5) = P(0 \leq X \leq 0.5, 0 \leq Y \leq 1)$$

$$= \int_0^1 \int_0^{0.5} f(x, y) dx dy$$

$$= \int_0^1 \int_0^{0.5} \frac{2}{3}(x+2y) dx dy$$

$$= \frac{2}{3} \int_0^1 \left(\frac{x^2}{2} + 2xy \right) \Big|_0^{0.5} dy$$

$$= \frac{2}{3} \int_0^1 \left(\frac{0.5^2}{2} + (2)(0.5)(y) \right) dy$$

$$= \frac{2}{3} \left(\frac{y^2}{2} + \frac{y}{1} \right) \Big|_0^1$$

$$= \frac{2}{3} \times \frac{5}{8}$$

$$= 5/12$$

Problem #9

$$f(x, y, z) = \begin{cases} \frac{4xyz^2}{9} & , 0 < x, y < 1, 0 < z < 3, \\ 0 & \text{elsewhere.} \end{cases}$$

- (a) the joint marginal density function of Y & Z

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$$g(y, z) = \int_x f(x, y, z) dx$$

$$= \int_0^1 \frac{4xyz^2}{9} dx$$

$$= \left(\frac{2x^2yz^2}{9} \right) \Big|_0^1$$

$$= \frac{2yz^2 \cdot 1^2}{9} - \frac{2yz^2 \cdot 0^2}{9}$$

$$= \frac{2yz^2}{9}$$

(b) the marginal density of y

$$h(y) = \int_x \int_z f(x, y, z) dz dx$$

$$= \int_0^1 \int_0^3 \frac{4xyz^2}{9} dz dx$$

$$= \int_0^1 \left(\frac{4xyz^3}{27} \right) \Big|_0^3 dx$$

$$= \int_0^1 \left(\frac{4xy \cdot (3)^3}{27} - \frac{4xy(0)^3}{27} \right) dx$$

$$= \int_0^1 4xy dx$$

$$= (2x^2y) \Big|_0^1$$

$$= 2y$$

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(c) $P\left(\frac{1}{4} < X < \frac{1}{2}, Y > \frac{1}{3}, 1 < Z < 2\right);$

$$P\left(\frac{1}{4} < X < \frac{1}{2}, Y > \frac{1}{3}, 1 < Z < 2\right) = \int_{1/4}^{1/2} \int_{1/3}^{\infty} \int_1^2 f(x, y, z) dz dy dx$$

$$= \int_{1/4}^{1/2} \int_{1/3}^1 \int_1^2 \frac{4xyz^2}{9} dz dy dx$$

$$= \int_{1/4}^{1/2} \int_{1/3}^1 \left(\frac{4xyz^3}{27} \right) \Big|_1^2 dy dx$$

$$= \int_{1/4}^{1/2} \int_{1/3}^1 \left(\frac{4xy \cdot 2^3}{27} - \frac{4xy \cdot 1^3}{27} \right) dy dx$$

$$= \int_{1/4}^{1/2} \int_{1/3}^1 \frac{28xy}{27} dy dx$$

$$= \int_{1/4}^{1/2} \left(\frac{14x \cdot (1)^2}{27} - \frac{14x \cdot (1/3)^2}{27} \right) dx$$

$$= \int_{1/4}^{1/2} \left(\frac{14x}{27} - \frac{14x}{243} \right) dx$$

$$= \int_{1/4}^{1/2} \frac{112x}{243} dx$$

$$= \left(\frac{56x^2}{243} \right) \Big|_{1/4}^{1/2}$$

$$= \frac{(56)(1/2)^2}{243} - \frac{56 \cdot (1/4)^2}{243}$$

$$= \frac{14}{243} - \frac{7}{486}$$

$$= \frac{7}{162}$$

(d) $P(0 < X < 1/2 \mid Y = 1/4, Z = 2)$

$$P(0 < X < \frac{1}{2} \mid Y = \frac{1}{4}, Z = 2) = \int_0^{1/2} f(x \mid \frac{1}{4}, 2) dx$$

$$= \int_0^{1/2} \frac{f(x, 1/4, 2)}{g(1/4, 2)} dx$$

$$= \int_0^{1/2} \frac{f(x, 1/4, 2)}{g(1/4, 2)} \cdot dx$$

$$\int_0^{1/2} \frac{4x \cdot \frac{1}{4} \cdot 2^3}{\frac{2 \cdot \frac{1}{4} \cdot 2^3}{9}} \cdot dx \quad \text{--- (1)}$$

$$= \int_0^{1/2} 2x dx$$

$$= (x^2) \Big|_0^{1/2}$$

$$= (1/2)^2 - 0^2$$

$$= 1/4$$

In (1) we used the formula for the joint marginal density function of Y & Z .

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Problem # 10

$$f(x, y) = \frac{9}{16} \cdot \frac{1}{4^{(x+y)}} \quad (x, y = 0, 1, 2, \dots)$$

$$x = 0, 1, 2, \dots \quad \text{and} \quad y = 0, 1, 2, \dots$$

- (a) Are the two random variables X & Y independent?
Explain why or why not.

$$f(x, y) = g(x) h(y).$$

$$g(x) = \sum_y f(x, y)$$

$$= \sum_{y=0}^{\infty} \frac{9}{16} \cdot \frac{1}{4^{x+y}}$$

$$= \frac{9}{16} \cdot \frac{1}{4^x} \sum_{y=0}^{\infty} \frac{1}{4^y}$$

$$= \frac{9}{16 \cdot 4^x} \sum_{y=0}^{\infty} \left(\frac{1}{4}\right)^y$$

$$= \frac{9}{16 \cdot 4^x} \cdot \frac{1}{1 - 1/4}$$

$$= \frac{3}{4^{x+1}}$$

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Problem # 11

Plot the following discrete probability function using x, y & $z = f(x, y)$ axes.

Solution:-

The total number of equally likely ways of selecting any point from the 8 r.s (8) = 28. The number of ways of selecting any $(2)^2$ red plans and 1 given from 3 green point is $\left(\frac{2}{1}\right)\left(\frac{3}{1}\right) = 6$.

$$\text{Hence } f(0,1) = \frac{6}{28} = \frac{3}{14}$$

The joint discrete probability function can be represented by the formula

$$f(x,y) = \frac{(3/x)(2/y)(2-x^3-y)}{(8/2)}$$

for $x = 0, 1, 2$ $y = 0, 1, 2$ & $0 \leq x+y \leq 2$

		x		
		0	1	2
y	0	3/28	9/28	3/28
	1	3/14	3/14	0
	2	1/28	0	0

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Now, we are able to check condition (i)

$$g(x)h(y) = \frac{3}{4^{x+1}} \cdot \frac{3}{4^{y+1}}$$

$$= \frac{9}{4^{(x+1)+(y+1)}}$$

$$= \frac{9}{16} \cdot \frac{1}{4^{x+y}}$$

$$= f(x, y)$$

Thus, the random variables X & Y are independent.

(b) What is the probability that during the time period in question less than 4 vehicles arrive at the two street owners?

$$P(X+Y < 4) = P(X+Y=0) + P(X+Y=1) + P(X+Y=2) + P(X+Y=3)$$

$$= P(X=0, Y=0) + (P(X=1, Y=0) + P(X=0, Y=1)) +$$

$$(P(X=2, Y=0) + P(X=1, Y=1) + P(X=0, Y=2))$$

$$+ (P(X=3, Y=0) + P(X=2, Y=1) + P(X=1, Y=2) + P(X=0, Y=3))$$

$$= f(0,0) + (f(1,0) + f(0,1)) + (f(2,0) + f(1,1) + f(0,2))$$

$$+ (f(3,0) + f(2,1) + f(1,2) + f(0,3)).$$

$$= \frac{9}{16} \left(\frac{1}{4^{0+0}} + \frac{1}{4^{1+0}} + \frac{1}{4^{0+1}} + \frac{1}{4^{2+0}} + \frac{1}{4^{1+1}} + \frac{1}{4^{0+2}} + \right.$$

$$\left. \frac{1}{4^{0+3}} + \frac{1}{4^{3+0}} + \frac{1}{4^{2+1}} + \frac{1}{4^{1+2}} + \frac{1}{4^{0+3}} \right)$$

$$= \frac{9}{16} \left(1 + 2 \cdot \frac{1}{4} + 3 \cdot \frac{1}{16} + 4 \cdot \frac{1}{64} \right)$$

$$= 63/64$$

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Problem #12

$$f(x) = \begin{cases} x & , 0 < x < 1 \\ 2-x & , 1 \leq x < 2 \\ 0 & \text{elsewhere} \end{cases}$$

Find the average number of hours per year that families run their vacuum cleaners

Solution:-

$$\begin{aligned} \mu &= E(X) = \int_{-\infty}^{\infty} x f(x) dx = \int_0^1 x \cdot x dx + \int_1^2 x \cdot (2-x) dx \\ &= \int_0^1 x^2 dx + \int_1^2 (2x - x^2) dx = \int_0^1 x^2 dx + \int_1^2 2x dx - \int_1^2 x^2 dx \\ &= \left. \frac{x^3}{3} \right|_0^1 + 2 \cdot \left. \frac{x^2}{2} \right|_1^2 - \left. \frac{x^3}{3} \right|_1^2 \\ &= \frac{1}{3} + 2 \left(\frac{4}{2} - \frac{1}{2} \right) - \left(\frac{8}{3} - \frac{1}{3} \right) \\ &= \frac{1}{3} + 3 - \frac{7}{3} = \frac{1+9-7}{3} = 1 \end{aligned}$$

So, the average number of hour per year that families run their vacuum cleaners is

$$(100)(\mu) = (100)(1) = 100 \text{ hours.}$$

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Problem # 13

Let X be a random variable with the following probability distribution

x	-3	6	9
$f(x)$	$1/6$	$1/2$	$1/3$

Find $\mu_g(X)$, where $g(x) = (2x+1)^2$

Solution:-

$$\mu_g(X) = \sum_x g(x) f(x)$$

where $f(x)$ is the probability distribution function of X . Using that, we obtain.

$$\mu_g(X) = \sum_x g(x) f(x)$$

$$= \sum_x (2x+1)^2 f(x)$$

$$= [2(-3)+1]^2 \cdot f(-3) + (2 \cdot 6 + 1)^2 \cdot f(6) + (2 \cdot 9 + 1)^2 \cdot f(9)$$

$$= 25 \cdot \frac{1}{6} + 100 \cdot \frac{1}{2} + 25 \cdot \frac{36}{3}$$

$$= 209$$

x	-3	6	9
$f(x)$	$1/6$	$1/2$	$1/3$
$g(x)$	25	100	361

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Problem # 14 $Y = X + 4$, where X has density function

$$f_X = \begin{cases} \frac{32}{(x+4)^3}, & x > 0 \\ 0, & \text{elsewhere} \end{cases}$$

Find average number of days that a person is hospitalized following treatment for this disorder.

Solution:-

$$E(Y) = E(X+4)$$

$$= \int_{-\infty}^{\infty} (x+4) f(x) dx$$

$$= \int_{x=0}^{\infty} (x+4) \cdot \frac{32}{(x+4)^3} dx$$

$$= \lim_{t \rightarrow \infty} \int_{x=0}^t \frac{32}{(x+4)^2} dx$$

$$= \lim_{t \rightarrow \infty} \left(-\frac{32}{x+4} \right) \Big|_0^t$$

$$= \lim_{t \rightarrow \infty} \left(-\frac{32}{t+4} + \frac{32}{0+4} \right)$$

$$= 8$$

The expected hospitalization period is 8 days.

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Problem #15

$f(x,y)$		x	
		2	4
y	1	0.10	0.15
	3	0.20	0.30
	5	0.10	0.15

Solution:-

- (a) Find the unexpected value of $g(X,Y) = XY^2$

$$\mu_g(X,Y) = E(g(X,Y)) = \sum_x \sum_y g(x,y)f(x,y)$$

$$\begin{aligned} &= g(2,1)f(2,1) + g(2,3)f(2,3) + g(2,5)f(2,5) \\ &\quad + g(4,1)f(4,1) + g(4,3)f(4,3) + g(4,5)f(4,5) \\ &= (2)(1^2)f(2,1) + (2)(3^2)f(2,3) + (2)(5^2)f(2,5) \\ &\quad + (4)(1^2)f(4,1) + (4)(3^2)f(4,3) + (4)(5^2)f(4,5) \\ &= (2)(1)(0.1) + (2)(9)(0.2) + (2)(25)(0.4) + (4)(1)(0.15) \\ &\quad + (4)(9)(0.3) + (4)(25)(0.15) \\ &= 0.2 + 3.6 + 5 + 0.6 + 10.8 + 15 \\ &= 35.2 \end{aligned}$$

so, the expected value of $g(X,Y)$ is $E(g(X,Y)) = 35.2$

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(b) Find μ_x & μ_y

Problem #16

Let $f_x(x) = P(X=x)$ & $f_y(y) = P(Y=y)$

$f(x,y)$		x		$f_y(y)$
		2	4	
y	1	0.1	0.15	0.25
	3	0.2	0.3	0.5
	5	0.1	0.15	0.25
$f_x(x)$		0.4	0.6	

$$\begin{aligned}\mu_x = E(X) &= \sum_x x f_x(x) = (2)(f_x(2)) + (4)(f_x(4)) \\ &= (2)(0.4) + (4)(0.6) \\ &= 0.8 + 2.4 \\ &= 3.2\end{aligned}$$

The expected value of discrete random variable Y is

$$\begin{aligned}\mu_y = E(Y) &= \sum_y y f_y(y) = (1)(f_y(1)) + (3)(f_y(3)) + (5)(f_y(5)) \\ &= (1)(0.25) + (3)(0.5) + (5)(0.25) \\ &= 0.25 + 1.5 + 1.25 \\ &= 3\end{aligned}$$

Problem #16

$$f(x,y) = \begin{cases} 4xy, & 0 < x, y < 1 \\ 0, & \text{elsewhere} \end{cases}$$

Find expected value of $z = \sqrt{x^2 + y^2}$

Solution:-

x & y are continuous random variables, so we are using the formula

$$\begin{aligned} E(Z) &= E(\sqrt{x^2 + y^2}) \\ &= \int_x \int_y \sqrt{x^2 + y^2} \cdot f(x,y) dy dx \\ &= \int_0^1 \int_0^1 \sqrt{x^2 + y^2} \cdot 4xy dy dx \end{aligned}$$

In order to compute the inner integral, let us use the substitution

$$t^2 = x^2 + y^2 \quad \text{--- (1)}$$

From there we obtain the following inequalities.

$$y = \sqrt{t^2 - x^2} \quad \text{--- (2)}$$

$$t = \sqrt{x^2 + y^2} \quad \text{--- (3)}$$

$$2t dt = 2y dy \rightarrow y dy = t dt \quad \text{--- (4)}$$

(2) is obtained by differentiating (1) with respect to y .

Now, we can continue the process

$$\begin{aligned} E(Z) &= \int_0^1 \int_0^1 \sqrt{x^2 + y^2} \cdot 4xy dy dx \\ &= \int_0^1 \int_0^1 4x \sqrt{x^2 + y^2} \cdot (y dy) dx \end{aligned}$$

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$$\begin{aligned}
&= \int_0^1 \left(\int_1^{\sqrt{x^2+1}} 4xt \cdot t dt \right) dx \\
&= \int_0^1 4x \left(\int_1^{\sqrt{x^2+1}} t^2 dt \right) dx \\
&= \int_0^1 4x \left(\frac{(\sqrt{x^2+1})^3}{3} \right) dx \\
&= \int_0^1 \left(\frac{4}{3} x(x^2+1)^{3/2} - \frac{4}{3} x^4 \right) dx \\
&= \frac{4}{3} \int_0^1 x(x^2+1)^{3/2} dx - \frac{4}{3} \int_0^1 x^4 dx
\end{aligned}$$

Let us solve the second integral from (3)

$$\begin{aligned}
\frac{4}{3} \int_0^1 x^4 dx &= \frac{4}{3} \left(\frac{1}{5} x^5 \right) \Big|_0^1 \\
&= \frac{4}{3} \cdot \frac{1}{5} \\
&= \frac{4}{15}
\end{aligned}$$

Now, let us solve the first integral from (3). We will again use a substitution this time

$$u^2 = x^2 + 1. \quad \text{--- (4)}$$

Thus, we obtain the following equalities

$$\begin{aligned}
x &= \sqrt{u^2 - 1} \\
u &= \sqrt{x^2 + 1}
\end{aligned}$$

$$2u du = 2x dx \rightarrow x dx = u du \quad \text{--- (5)}$$

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we obtained (5) by differentiation of (4) with respect to a .

Now, we are able to compute the first integral in (3).

$$\frac{4}{3} \int_0^2 x(x^2+1)^{\frac{3}{2}} dx = \frac{4}{3} \int_1^5 (\sqrt{x^2+1})^3 \cdot (x dx)$$

$$= \frac{4}{3} \int_1^5 u^3 \cdot (udu).$$

$$= \frac{4}{3} \int_1^5 u^4 du.$$

$$= \frac{4}{3} \left(\frac{1}{5} (\sqrt{x^2+1})^5 \right) \Big|_1^5$$

$$= \frac{4}{3} \left(\frac{(\sqrt{2})^5}{5} - \frac{1}{5} \right).$$

$$= \frac{16\sqrt{2}}{15} - \frac{4}{15}$$

By inserting these obtained values back into (3) we get.

$$E(Z) = \frac{16\sqrt{2}}{15} - 8$$

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Problem # 17

The random variable X , representing the number of errors per 100 lines of software code, has the following distribution.

x	2	3	4	5	6
$f(x)$	0.01	0.25	0.4	0.3	0.04

Find the variance of X .

Solution:-

We will determine the variance of discrete random variable X by using the statement.

$$\sigma^2_X = E(X^2) - \mu^2_X$$

In order to do so, we first need to determine the mean of X .

$$\mu_X = \sum_x x f(x)$$

$$= \sum_{x=2}^6 x f(x)$$

$$= 2 \times 0.01 + 3 \times 0.25 + 4 \times 0.4 + 5 \times 0.3 + 6 \times 0.04$$

$$= 4.11$$

Further on, let's find the expected value of X^2 .

$$E(X^2) = \sum_x x^2 f(x)$$

$$= \sum_{x=2}^6 x^2 f(x)$$

$$= (2)^2(0.01) + (3)^2(0.25) + (4)^2(0.4) + (5)^2(0.3)$$

$$+ (6)^2(0.04)$$

$$= 17.63$$

$$\sigma^2_X = E(X^2) - \mu^2_X = 17.63 - 4.11^2 = 0.7379$$

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Problem # 18

Random variable X & Y follow a joint distribution

$$f(x, y) = \begin{cases} 2 & 0 < x \leq y < 1 \\ 0 & \text{otherwise} \end{cases}$$

Determine the correlation coefficient between X & Y

Solution:-

$$\rho_{XY} = \frac{\sigma_{XY}}{\sigma_X \sigma_Y}$$

$$\sigma_X^2 = E(X^2) - \mu^2_X$$

$$= \int_x \int_y x^2 f(x, y) - \left(\int_x \int_y x f(x, y) \right)^2$$

$$= \int_0^1 \int_x^1 x^2 \cdot 2 dy dx - \left(\int_0^1 \int_x^1 x \cdot 2 dy dx \right)^2$$

$$= \int_0^1 \left(2x^2 y \right) \Big|_x^1 dx - \left(\int_0^1 \left(2xy \right) \Big|_x^1 dx \right)^2$$

$$= \int_0^1 (2x^2 - 2x^3) dx - \left(\int_0^1 (2x - 2x^2) dx \right)^2$$

$$= \left(\frac{2}{3} x^3 - \frac{1}{2} x^4 \right) \Big|_0^1 - \left(\left(x^2 - \frac{2}{3} x^3 \right) \Big|_0^1 \right)^2$$

$$= \frac{2}{3} - \frac{1}{2} \left(1 - \frac{2}{3} \right)^2 = \frac{1}{6} - \frac{1}{9}$$

$$= \frac{1}{18}$$

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$$\begin{aligned}
\sigma^2_y &= E(Y^2) - \mu^2_y \\
&= \int_x \int_y y^2 f(x, y) - \left(\int_x \int_y y f(x, y) \right)^2 \\
&= \int_0^1 \int_x y^2 \cdot 2 dy dx - \left(\int_0^1 \int_x y \cdot 2 dy dx \right)^2 \\
&= \int_0^1 \left(\frac{2}{3} y^3 \right) \Big|_x^1 dx - \left(\int_0^1 (y^2) \Big|_x^1 dx \right)^2 \\
&= \int_0^1 \left(\frac{2}{3} - \frac{2}{3} x^3 \right) dx - \left(\int_0^1 (1 - x^2) dx \right)^2 \\
&= \left(\frac{2}{3} x - \frac{1}{6} x^4 \right) \Big|_0^1 - \left(\left(x - \frac{1}{3} x^3 \right) \Big|_0^1 \right)^2 \\
&= \int_0^1 \left(\frac{2}{3} - \frac{2}{3} x^3 \right) dx - \left(\int_0^1 (1 - x^2) dx \right)^2 \\
&= \left(\frac{2}{3} x - \frac{1}{6} x^4 \right) \Big|_0^1 - \left(\left(x - \frac{1}{3} x^3 \right) \Big|_0^1 \right)^2 \\
&= \frac{2}{3} - \frac{1}{6} - \left(1 - \frac{1}{3} \right)^2 \\
&= \frac{1}{2} - \frac{4}{9} \\
&= \frac{1}{18}
\end{aligned}$$

Next we need to find the covariance of X & Y .
 From calculating the expected values we know that
 $\mu_x = \frac{1}{3}$ & $\mu_y = \frac{2}{3}$

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$$\begin{aligned}\sigma_{xy} &= E(XY) - \mu_x \mu_y \\&= \int_x \int_y xy f(x, y) - \frac{1}{3} \cdot \frac{2}{3} \\&= \int_0^1 \int_x xy \cdot 2 dy dx - \frac{2}{9} \\&= \int_0^1 (xy^2) \Big|_x^1 dx - \frac{2}{9} \\&= \int_0^1 (x - x^3) dx - \frac{2}{9} \\&= \left(\frac{1}{2} x^2 - \frac{1}{4} x^4 \right) \Big|_0^1 - \frac{2}{9} \\&= \frac{1}{2} - \frac{1}{4} - \frac{2}{9} \\&= \frac{1}{36}\end{aligned}$$

Finally, by using formula, we obtain

$$\rho_{xy} = \frac{\sigma_{xy}}{\sigma_x \sigma_y} = \frac{1/36}{\sqrt{1/18} \cdot \sqrt{1/18}}$$

$$\rho_{xy} = \frac{1}{36} / \frac{1}{18}$$

$$\rho_{xy} = \frac{1}{2}$$